

House of Trae Infrastructure Architecture Specification

Version 3.0

Executive Summary

The House of Trae Infrastructure Platform is a self-hosted, security-focused operating environment designed to support multiple business entities, client systems, personal services, and future software products from a unified architecture.

The platform is built upon four core pillars:

1. Security by Design
2. Operational Simplicity
3. Service Modularity
4. Ownership of Data and Infrastructure

The architecture is centred around Proxmox virtualization, Keycloak identity management, Caddy edge routing, PowerDNS authoritative DNS, and the PrivateNexus operations platform.

Strategic Objectives

Business Objectives

- Support multiple business brands.
- Support customer-facing applications.
- Support internal operational tooling.
- Provide centralized management.
- Reduce infrastructure sprawl.

Technical Objectives

- Centralized authentication.
- Standardized deployments.
- Repeatable recovery procedures.
- Full observability.
- Strong security boundaries.

Core Platform Components

Infrastructure Layer

- Proxmox VE
- ZFS Storage
- VLAN Segmentation
- Tailscale Overlay Network

Identity Layer

- Keycloak
- MFA Enforcement
- RBAC

Gateway Layer

- Caddy
- TLS Automation
- Reverse Proxy
- Rate Limiting

Operations Layer

- PrivateNexus
- Monitoring
- Backup Visibility
- Recovery Planning

Service Layer

- ERP Systems
- Collaboration Platforms
- Personal Cloud Services
- Business Websites

Success Criteria

The platform shall provide a secure, resilient, and maintainable environment capable of supporting both current requirements and future business growth without requiring architectural redesign.